Science By doing

1. Go to 4.1, “notebook” and review vocabulary by clicking on ‘hydrological cycle’.
2. Under 4.2, what are the six ways that the ‘urban water cycle’ differs to the natural water cycle?

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3. Under 4.3, list the five steps in water filtration and write one sentence summarising each process.

4. Read through the information on Science By Doing 4.4 and see if there are any noticeable differences in the water. If there are, write them in this table.

Characteristic	Mpwelarra water	Katoomba water	Possible explanation?

5. **<u>What does an aquatic ecologist measure?</u>** Under 4.5 go to “find out more” and fill in the blanks

Dissolved oxygen

Measures the amount of oxygen in the water. Very high or very low levels can affect organisms and indicate an unbalanced or unhealthy waterway.

pH

Measures _____/_____of the water. Highly acidic or alkaline waters can be toxic to aquatic organisms.

Turbidity and suspended solids

Turbidity is a measure of the _____ of the waterway. Total suspended solids measure the amount of particles _____ in the water. Small suspended particles of soil and organic matter are naturally present in all waters. In high concentration they reduce the _____ of light (which is necessary for plants), or the particles can settle and smother aquatic organisms.

Nitrogen and Phosphorous

These are among the most important nutrients for plant growth. Concentrations are generally _____ in natural waterways, but are dramatically increased by some human activities. Very high concentrations of nutrients cause excessive plant growth and an increased spread of weeds and _____.

Chlorophyll-a

This is a green pigment found in most plants including _____. It is an indicator of micro algal _____. High concentrations occur in algal blooms which indicate poor ecosystem health.

Macro invertebrates

The presence or not of macro invertebrates gives an indication of water quality.

Extension: Use the following table to answer the questions

Water samples from Cascade water filtration plant (Katoomba catchment) and Mpwelarre (Walkabout Bore)

Testing	Characteristics	Measuring units	Typical average Katoomba sample	Typical average Mpwelarre sample	Australian Drinking Water Guidelines	
					Health	Aesthetics
Microbiological	E. coli	Cfu/100mL	<1	<1	0	
Physical	Turbidity	NTU	0.08	1		5
	True colour	HU	<2	Not tested		<15
	pH	pH units	7.44	7.3		6.5–8.5
	Total hardness	ppm	57	382		200
	Total dissolved solids	mg/L	119	1010		<500
	Conductivity	mS/m	14	169	NA	NA
Chemical	Free chlorine	mg/L	0.54	NA	<5	<0.6
	Monochloramine	mg/L	0.09	NA	<3	<0.5
	Fluoride	mg/L	0.99	0.9	0.9–1.5	
	Total Trihalomethanes (disinfectant by product)	mg/L	0.072	ND	<0.25	
	Chloride	mg/L	20.75	200		<250
	Iron	mg/L	0.022	0.34		<0.3
	Aluminium	mg/L	0.0215	0.02		<0.2
	Manganese	mg/L	0.005	0.05	<0.5	<0.1
	Arsenic	mg/L	ND	0.001	<0.007	
	Copper	mg/L	0.023	0.01	<2	<1
	Mercury	mg/L	ND	0.0002	<0.001	
	Nitrate	mg/L	0.29	7.3	<50	
	Nitrite	mg/L	0.003	<0.005	<3	
	Selenium	mg/L	ND	0.008	<0.01	
	Sodium	mg/L	9.45	188		<180
	Iodide	mg/L	NA	0.18	<0.2	
Radiological	Uranium	mg/L	NA	0.009	<0.02	
	Radium 226 *	Bq/L	NA	0.105	5	
	Radium 228 *	Bq/L	NA	0.424	5	

1. Circle the values which exceed the Australian Water Drinking Guidelines.
2. Which water is healthier and what might be some reasons why?

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3. Choose three of these values to investigate further and research why they have a guideline in place (i.e. what are the health impacts if it is not controlled).

Water quality measure	Why it is controlled